

Advanced NMAP

Part -3

The earlier articles on Nmap had covered the basic features, the working of the TCP 3-way handshake, and some important scan types, along with their practical use. This article continues with some more interesting scanning techniques.

A very important aspect of Nmap scanning is the Nmap ping process, sometimes also called the host discovery/ping scan process. As documented on the nmap.org site, an Nmap scan of a subnet usually begins with discovering which hosts in the subnet are online. This host discovery is in operation every time any of the Nmap scan types is run. It is different from standard ICMP pings, and combines ARP requests and elaborate combinations of TCP, ICMP and other probes. The exact type of scan is decided based on whether the Nmap workstation is scanning its own (local) subnet, or an external subnet. Host discovery is followed by deeper investigation of the online hosts.

Typical Nmap host discovery

Local subnets

For scanning devices in a local subnet, Nmap uses an ARP scan, where an ARP request is sent to the local device. The device acknowledges this with an ARP reply, thus revealing its

presence. Being at the OSI Layer 2, ARP requests are limited to only the local subnet.

Remote subnets

To scan remote subnets, Nmap sends an ICMP echo packet and a TCP ACK packet to the remote device. The remote device may send an ICMP reply and TCP RST respectively, thus revealing its presence.

Assume online switch

Now, consider a real-life scenario where all hosts on the local network have personal firewalls, and the default ping scan shows no active hosts. How do you probe this network further? Nmap provides an excellent option `-PN`, which tells Nmap not to do a ping scan to discover active hosts, but to assume that all hosts in the range being scanned are online.

The implications of disabling host discovery for even a Class C subnet with 254 IP addresses is that all the 254 IP addresses will be probed, including non-existent hosts, thus increasing scan time to a great extent. Exercise care in using this option. For example, `nmap -v -PN 192.168.100.0/24` will scan